

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Pharmacy SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Pharmacy Branch : Pharmacy

Semester : SEMESTER - 8

Subject Code & Name: BP801T - BIostatistics & Research Methodology

Time : 3 Hours]

[Total Marks : 75

Instructions to the Students:

- 1.All questions are compulsory.
- 2.Draw diagrams / figures wherever necessary
- 3.Figures to right indicate full marks

Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

- | | |
|--|---|
| A) What is blocking and confounding effect in factorials? | 2 |
| B) Differentiate between parametric and non-parametric tests. | 2 |
| C) Compare how we can differentiate Type I Error and Type II Error | 2 |
| D) Define Cohort studies and observational study. | 2 |
| E) Summarize the use MS Excel in Pharma industry | 2 |
| F) Reframe outline of Research Protocol | 2 |
| G) What is concept of Blocking and co-founding system? | 2 |
| H) What is Regression? | 2 |
| I) Explain the correlation and causation effect. | 2 |
| J) Justify need of design for experiment. | 2 |

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

- | | |
|--|----|
| A) What is Research? Write a note on Experimental Design Techniques | 10 |
| B) What is frequency distribution? explain all measures of central tendency and measures of dispersion with example. | 10 |
| C) Define Standard deviation, Write Application of it. Calculate the Mean and Standard deviation of tablets not passes test of disintegration which are collected from 10 different container Number of tablets (Not passes test) :
5,6,8,7,5,3,4,3,6,3 | 10 |

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

- | | |
|--|---|
| A) Recall Practical components used in Industry. Write note on Minitab and SPSS. | 5 |
| B) Write note on Histogram and Pie Chart | 5 |
| C) Write note on simple regression model | 5 |
| D) Write detail account on Parametric test | 5 |
| E) What is sampling, Explain in details types of sampling. | 5 |
| F) Define Correlation: Justify Karl Pearson's coefficient of correlation | 5 |
| G) Define of Probability, Explain Binomial distribution, and Normal distribution | 5 |
| H) Describe report writing procedures and presentation of data. | 5 |
| I) What are distribution free tests, explain sign test and Wilcoxon signed rank test | 5 |

*** End ***

068275752

068275752

068275752

068275752

068275752

068275752

068275752

068275752

068275752